# AI-Enabled Operational & Logistics Transformation at NAC Kazatomprom JSC

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# Kazatomprom should run three AI pilots, with people keeping control of safety and finance

*Pilot SCADA telemetry and AI acid dosing at Inkai, adapt AI freight tracking on the TITR, and pilot one robotic packing arm at Ulba, while emergency shutdowns, groundwater protection and financial checks stay human-controlled [1], [7].*

**1**

**Inkai: 90-day SCADA pilot**

Proposed target: 5%–6% less acid per kg U

**2**

**TITR: AI freight tracking**

Proposed target: 10 days shorter transit

**3**

**Ulba: drum-packing pilot**

Proposed target: 30% fewer exposure hours

# Acid supply, corridor delays and locked-up inventory limit Kazatomprom more than demand does

- Sulfuric acid shortage led to lower 2025 production expectations [8]
- About 48% of Western shipments moved to the TITR (Assignment 1 evidence); adds transfers, Caspian weather and customs [9]
- KZT 423,314M inventory at end-2023, KZT 328,015M of it uranium products [1]
- Result: the binding limits are chemistry, transit time and cash, not sales

Sources: [1], [8], [9]

# The state owner wants domestic investment, and the Subsoil Code limits how fast AI can change operations

| Stakeholder | What they expect | Effect on my design |
|---|---|---|
| Samruk-Kazyna (75%) | Dividends, capital kept in Kazakhstan, regional jobs | Reject foreign M&A; retrain staff |
| Public investors (25%) | Returns, clear reporting (London, Astana) | Named reviewer for all financial figures |
| Ministry of Energy | Subsoil Code groundwater protection | Shutdowns and escalations stay human-only |

Sources: [1], [7]

# Protect low-cost ISR extraction and add software, not foreign reactor assets

- Kazatomprom: 100% in-situ recovery; acid solution through sandy ground, so no ore is moved [6]
- Cameco: underground hard-rock mining (McArthur River, Cigar Lake) and a 49% Westinghouse stake [3]
- Different mining method means only process software and logistics practices can transfer
- Foreign reactor M&A conflicts with the 75% state owner’s domestic focus [1]

Sources: [1], [3], [6]

# Reject two Cameco ideas, adopt or pilot two, and adapt one

| Cameco idea | Verdict | Why |
|---|---|---|
| 49% Westinghouse-style M&A [3] | Reject | Conflicts with state capital goals [1] |
| Underground mining automation [3] | Reject | ISR has no hard-rock mines [6] |
| Automated process control, telemetry | Adopt / Pilot | Fits ISR; targets acid use [6] |
| Automated drum packing | Pilot | One arm at Ulba |
| Shipment tracking after Inkai delay [3] | Adapt | Redesign for multi-border TITR [9] |

Sources: [3], [6], [9]

# AI screens and drafts; named people decide on safety, law and money

| Tier | Wellfield | Ulba | TITR | Finance |
|---|---|---|---|---|
| Human only | Shutdowns, groundwater | Safety, licence | Contract talks | Sign statements |
| Human + AI | Reserve modelling | Pack scheduling | Inventory swaps | Scenarios |
| AI + human review | Dosing alerts; Lead Engineer signs above ±5% | Fault alerts; supervisor | ETA variance >24 h; Logistics Officer | Ratios; Senior Analyst |
| Automated | Telemetry, 20,000+ wellheads | Drum filling | Status data | Data matching |
| Fallback | >3% drift: 6-h manual rounds | Manual tele-manipulator | Manual customs filing | Manual check |

Sources: [7], [10] (proposed design)

# Continuous telemetry would replace a 48–72 hour manual lag (design assumption)

| | Current state (assumed) | Proposed 90-day pilot, one Inkai block |
|---|---|---|
| Sampling | Manual rounds, 20,000+ wellheads | Continuous SCADA downhole telemetry |
| Delay | 48–72 h lab results | Near real time |
| Dosing | Late valve changes | AI drafts; Lead Engineer approves shifts above ±5% |
| Escalation | — | >12 bar alerts Environmental Safety Director |
| Fallback | — | Drift >3%: manual 6-hour rounds |
| Target | — | 5%–6% less acid per kg U (proposed) |

Sources: [6]; parameters proposed in Assignments 1 and 2

# Utilities buy reliability, so tracking and buffers matter as much as price

- Cameco 2023: only two-thirds of an Inkai shipment arrived in Q4; the rest came later at a Canadian port. Managed with inventory buffers, long-term contracts and loans [3]
- Kazatomprom has an ANU Energy inventory loan arrangement [1]
- Proposed TITR tracking: AI predicts ETA; Logistics Officer checks variance >24 h; port closure >72 h goes to Commercial Director
- Manual customs filing if the data feed fails
- Proposed target: 10 days shorter European transit [9]

Sources: [1], [3], [9]

# Staff need judgement, adaptability and the skill to check AI

| Capability | Example at Kazatomprom | Who is trained |
|---|---|---|
| Human | Ethical judgement on delays and groundwater risk | Wellfield engineers |
| Meta | Adapting when a route closes | Logistics staff |
| Digital | Checking AI ETAs, dosing alerts and ratios | Engineers, junior analysts |

*WEF 2025 [4] and OECD 2024 [5] discuss shifting skill demand; they are general evidence, not Kazatomprom data.*

Sources: [4], [5]

# Every AI action has a named approver, a limit and a manual fallback

| Risk | Control | Owner |
|---|---|---|
| Privacy, security | Data stays in company systems; role-based approval | IT Security Lead |
| Route bias | Check recommendations against alternate corridors | Logistics Officer |
| Telemetry reliability | Drift >3% → manual 6-hour hydrometer rounds | Lead Engineer |
| Accountability | Action log: who approved what | Operations Manager |
| Safety | >12 bar → Environmental Safety Director halts block | Env. Safety Director |
| Customs data loss | Manual customs filing | Logistics Officer |

Sources: [7] (proposed controls)

# Profit is strong, but only KZT 211,912M is cash against KZT 423,314M in inventory

| Audited 2023 (KZT million) | Value | Audited 2023 | Value |
|---|---|---|---|
| Revenue | 1,434,635 | Current liabilities | 430,812 |
| Net profit | 580,335 | Current ratio | 3.00 |
| Net profit margin | 40.45% | Inventory (uranium products) | 423,314 (328,015) |
| Current assets | 1,290,336 | Cash and equivalents | 211,912 |
| Operating cash flow | 432,225 | Cash ratio | 0.49 |

Sources: [1], [2]

# Only the 2023 figures are verified; pilot KPIs are proposed targets

| Item | Status | Basis |
|---|---|---|
| 2023 financials and ratios | Verified | Audited statements [1], [2] |
| Ownership; Cameco facts | Verified | [1], [3] |
| 48–72 h lag; 20,000+ wellheads; ±5%; 12 bar; 3%; 6 h | Design assumption | Assignments 1 and 2 |
| 5%–6% less acid; −10 days; −30% exposure | Pilot target | Benchmarked on [6], [3] |
| Pilot costs and KZT savings | Not estimated | No verified data |

Sources: [1], [2], [3], [6]

# A 12-month plan with owners, gates and fallbacks

| Action | Owner | KPI (baseline → target) | Dependency | Cost basis | Risk + control | Gate |
|---|---|---|---|---|---|---|
| M0–3: SCADA telemetry + ±5% dosing gate, one Inkai block | Inkai Ops Manager; Lead Engineer | Lag 48–72 h → near real time; acid → 5%–6% lower | Sensors, IT | Pilot sensors; quote needed | Drift; manual rounds | M3: scale or stop |
| M4–8: TITR tracking; Ulba robot pilot | Logistics Officer; Plant Safety Mgr | Transit → −10 days; exposure → −30% | Data access; safety approval | Licence, robot; quotes needed | API outage, robot fault; manual fallbacks | M8: extend or stop |
| M9–12: scale to more blocks; train staff; quarterly ratio checks | CTO; HR; Senior Analyst | Blocks 1 → agreed; zero unreconciled ratios | Pilot results | Training time | Over-trust in AI; review gate | M12: board decision |

Sources: Proposed plan; targets are pilot targets

# The model fits together: the plan runs only inside governance and cash limits




- §4 sets who approves; §6 sets the limits that §5 actions must stay within
- §7 shows cash is tight (0.49), so §8 gates each step on pilot results
- Any failed gate shrinks scope; it never removes human review

Note: diagram laid out with AI assistance (Claude) from my own analysis; no AI-generated images.

# I corrected an unverified 4.02 current ratio to 3.00 and declared AI use

| Check | Early draft | Audited result |
|---|---|---|
| Current assets | KZT 1,250.0bn (unverified) | KZT 1,290,336M |
| Current liabilities | KZT 310.5bn (unverified) | KZT 430,812M |
| Current ratio | 4.02 | 3.00 |
| Cash ratio | not calculated | 0.49 |

Level 3 AI use: ChatGPT (brainstorming), Gemini (cross-checking, mapping), Claude (website and drafting of report and deck). I checked all figures against [1], [2] and made the verdicts myself.

Sources: [1]

# If a shock hits, scope shrinks but human review stays

| Shock | Still holds | Changes | Abandoned |
|---|---|---|---|
| (a) Sensor drift or failure | Human shutdown rule; 12 bar escalation | Block returns to 6-h manual rounds | Automated dosing on that block until trusted |
| (b) TITR corridor or customs shock | 24 h variance check; 72 h escalation | Use inventory buffers [1]; manual customs | 10-day target for that period |
| (c) Engineer shortage | Human-only tier; ±5% gate | One block only; training earlier | Scale-up in Months 9–12 |

Sources: [1], [7]

# References


[1] NAC Kazatomprom JSC, Integrated Annual Report 2023, Astana, 2024.

[2] NAC Kazatomprom JSC, “Financial results of NAC Kazatomprom JSC for 2023,” Astana, 2024.

[3] Cameco Corporation, 2023 Annual Report (Form 40-F / Ex. 99.1), Saskatoon, 2024.

[4] World Economic Forum, The Future of Jobs Report 2025, Geneva, 2025.

[5] A. Green, “AI and the changing demand for skills,” OECD AI Papers, no. 14, Apr. 2024.

[6] IAEA, Manual of Acid In Situ Leach Uranium Mining Technology, IAEA-TECDOC-1239, Vienna, 2001.

[7] Republic of Kazakhstan, Code on Subsoil and Subsoil Use (No. 125-VI), Dec. 27, 2017.

[8] World Nuclear News, “Kazatomprom lowers 2025 uranium production expectations,” Aug. 2024.

[9] The Astana Times, “Kazakhstan Increases Uranium Shipments via TITR,” May 2024.

[10] G. Yeshenkulova et al., Business Administration, Almaty: ZIALY BASPASY, 2026; OpenStax, Introduction to Business, 2nd ed., 2026.